

252-0029-00L Parallel Programming

Semester	Spring Semester 2026
Lecturers	T. Hoefler, B. Solenthaler
Periodicity	yearly recurring course
Language of instruction	German

◀ Tabs

Courses

Number	Title	Hours	Lecturers
252-0029-00 V	Parallele Programmierung Findet im HG F 7 mit Videoübertragung ins HG F 5 statt.	4 hrs Mon 10:15-12:00 HG F 5 » 10:15-12:00 HG F 7 » Tue 10:15-12:00 HG F 5 » 10:15-12:00 HG F 7 »	T. Hoefler, B. Solenthaler
252-0029-00 U	Parallele Programmierung Groups are selected in myStudies.	2 hrs Wed 16:15-18:00 CHN D 48 » 16:15-18:00 CHN E 42 » 16:15-18:00 ETZ E 8 » 16:15-18:00 ETZ F 91 » 16:15-18:00 ETZ H 91 » 16:15-18:00 ETZ K 91 » 16:15-18:00 LFW C 1 » 16:15-18:00 LFW C 11 » 16:15-18:00 ML J 34.3 » Fri 10:15-12:00 CHN D 42 » 10:15-12:00 CLA E 4 » 10:15-12:00 HG G 26.5 » 10:15-12:00 IFW C 31 » 10:15-12:00 LFW B 2 » 10:15-12:00 LFW C 1 » 10:15-12:00 ML H 34.3 » 10:15-12:00 ML J 34.1 » 10:15-12:00 NO D 11 » 10:15-12:00 NO E 11 »	T. Hoefler, B. Solenthaler

Catalogue data

Abstract	Introduction to parallel programming: deterministic and non-deterministic programs, models for parallel computation, synchronization, communication, and fairness.
Learning objective	The student should learn how to write a correct parallel program, how to measure its efficiency, and how to reason about a parallel program. Student should become familiar with issues,

problems, pitfalls, and solutions related to the construction of parallel programs. Labs provide an opportunity to gain experience with threads, libraries for thread management in modern programming languages (e.g., Java, C#) and with the execution of parallel programs on multi-processor/multi-core computers.

Competencies 	Subject-specific Competencies	Concepts and Theories	assessed
		Techniques and Technologies	fostered
	Method-specific Competencies	Analytical Competencies	assessed
		Problem-solving	assessed
		Project Management	fostered

Performance assessment

Performance assessment information (valid until the course unit is held again)

► Performance assessment as a semester course

In examination block for Bachelor's Degree Programme in Computer Science 2016; Version 07.11.2024 (First Year Examination Block 2)

ECTS credits 7 credits

Examiners T. Hoefler, B. Solenthaler

Type session examination

Language of examination German

Repetition The performance assessment is offered every session. Repetition possible without re-enrolling for the course unit.

Mode of examination written 120 minutes

Written aids None

If the course unit is part of an examination block, the credits are allocated for the successful completion of the whole block.
This information can be updated until the beginning of the semester; information on the examination timetable is binding.

Learning materials

Main link **Information**

Only public learning materials are listed.

Groups

	252-0029-00 U	Parallele Programmierung
Groups	G-01 Fokusgruppe	Wed 16:15-18:00 ETZ E 8 »
	G-02	Wed 16:15-18:00 ETZ F 91 »
	G-03	Wed 16:15-18:00 CHN D 48 »
	G-04	Wed 16:15-18:00 CHN E 42 »
	G-05 Fokusgruppe	Fri 10:15-12:00 LFW B 2 »
	G-06	Wed 16:15-18:00 ETZ H 91 »
	G-07	Wed 16:15-18:00 LFW C 11 »

G-08 Fokusgruppe	Wed 16:15-18:00 ETZ K 91 »
G-09 Fokusgruppe	Wed 16:15-18:00 ML J 34.3 »
G-10	Wed 16:15-18:00 LFW C 1 »
G-11	Fri 10:15-12:00 HG G 26.5 »
G-12	Fri 10:15-12:00 ML J 34.1 »
G-13	Fri 10:15-12:00 NO D 11 »
G-14	Fri 10:15-12:00 CHN D 42 »
G-15 Fokusgruppe	Fri 10:15-12:00 LFW C 1 »
G-16	Fri 10:15-12:00 ML H 34.3 »
G-17 Fokusgruppe	Fri 10:15-12:00 CLA E 4 »
G-18	Fri 10:15-12:00 IFW C 31 »
G-19 Fokusgruppe	Fri 10:15-12:00 NO E 11 »

Restrictions

There are no additional restrictions for the registration.

Offered in

Programme	Section	Type	
Computer Science Bachelor	First Year Examination Block 2	O	